CALIFORNIA 2020 WILDFIRE ASSESSMENT

# From 4.2 million acres burned to the dollar cost of damage

*A five-act geospatial analysis :*

*statewide context → habitat & severity → the wildland-urban interface → built-asset valuation → what's next.*

Anna Mika · June 2026 · Environmental Risk Assessment · Albert School · Analysis in Python (GeoPandas · rasterio · rioxarray · Folium)

EXECUTIVE SUMMARY

# Headlines from the 2020 season

**504**

fires burned a 4.18M-acre footprint — 89% of acreage in Aug–Sep

**3.81M**

acres mapped for burn severity (MTBS); High severity = 0.73M ac

**28,626**

structures inspected (DINS) — 9,494 destroyed

**$1.8–2.6B**

built-asset loss (assessed-value lower bound, not total economic loss)

**Glass**

data-selected hotspot (Wine Country); Intermix WUI holds the most destroyed

DATA SOURCES

# Vector layers

| Layer | Source | Route | Rows | CRS |
|---|---|---|---|---|
| Fire perimeters 2020 | CAL FIRE FRAP — "California Fire Perimeters (all)" | CNRA open-data portal, filtered to 2020 | 504 | 3857 |
| DINS structure damage | CAL FIRE DINS (POSTFIRE_MASTER_DATA_SHARE) | ArcGIS FeatureServer, INCIDENTSTARTDATE in 2020 | 28,626 | 4326 |
| Wildland-Urban Interface | CAL FIRE / CNRA WUI (3 classes) | CA open-data portal GeoJSON | 36,157 | 4326 |
| County boundaries | US Census TIGER/Line 2020 (cartographic) | via pygris (CA counties) | 58 | 4269 |

DATA SOURCES

# Raster layers

## FVEG WHR13 vegetation

CAL FIRE FRAP — "Vegetation by Wildlife Habitat Relationships 2022" (updated with LANDFIRE 2020 EVT v2.2.0)

| | |
|---|---|
| **Format** | File geodatabase; single-band raster |
| **Classes** | Raster Attribute Table = 42,032 rows → 13-class WHR13NAME |
| **CRS / res** | EPSG:3310, 30 m, uint16 |

## MTBS burn severity

USGS / USFS Monitoring Trends in Burn Severity

| | |
|---|---|
| **Format** | CA 2020 annual thematic mosaic (single GeoTIFF) |
| **Classes** | 0–6, read from the .xml sidecar |
| **CRS / res** | EPSG:5070, 30 m, uint8, nodata = 0 |

MTBS class map: 0 = background/nodata · 1 = Unburned-Low · 2 = Low · 3 = Moderate · 4 = High · 5 = Increased Greenness · 6 = Non-Processing Mask.

METHODOLOGY

# The pipeline: a two-notebook, analysis-first workflow

## 01_data_prep

- load → inspect / schema-discovery
- clean → reproject
- spatial joins
- raster clips
- write data/processed/

## 02_analysis_viz

- reads only data/processed/
- Acts 1–5 of analysis
- figures, maps & animations
- written to outputs/

*Principle: heavy prep runs once · every intermediate persisted · visuals never re-run prep.*

# Seven preparation steps

### 1.1 Inspect / schema-discovery

Confirm field names, CRSs, raster attribute tables, severity-class definitions in-notebook — assume nothing.

### 1.2 Clean DINS

Ordered DAMAGE categorical; numeric assessed value; damage_fraction; value-coverage flags; geometry verified vs lat/lon.

### 1.3 Clean perimeters

Repair geometry; area cross-check; parse dates.

### 1.4 WUI prep

Dissolve by class (Interface / Intermix / Influence Zone).

### 1.5 Spatial joins

Structures → county / WUI / perimeter.

### 1.6 Raster clips

Clip FVEG & MTBS to buffered perimeters, then tabulate.

### 1.7 Write processed outputs

Persist cleaned layers + tabulations + manifest.

# Spatial-join coverage & the FIRENAME check

## Join coverage of 28,262 structures

**100%**
county

**85.2%**
WUI

**74.5%**
perimeter

- The 74.5% is ok — perimeter match by damage class:
  - Destroyed 99.1%
  - Major 97.2%
  - Minor 91.2%
  - Affected 91.8%
  - No Damage only 60.2%.
- Undamaged survivors sit on the fire margin; any-damage = 98.4% inside a perimeter.

## FIRENAME attribute check

**55.7%**
Perfect match

**7,208 of 16,261 comparable "mismatch" (44.3%) —> a naming-convention difference, not a spatial error.**

- BEAR → North Complex
- Napa/Sonoma/Santa Rosa → Glass

Spatial join is authoritative.

METHODOLOGY · ACT 4 PREVIEW

# The Valuation Approach

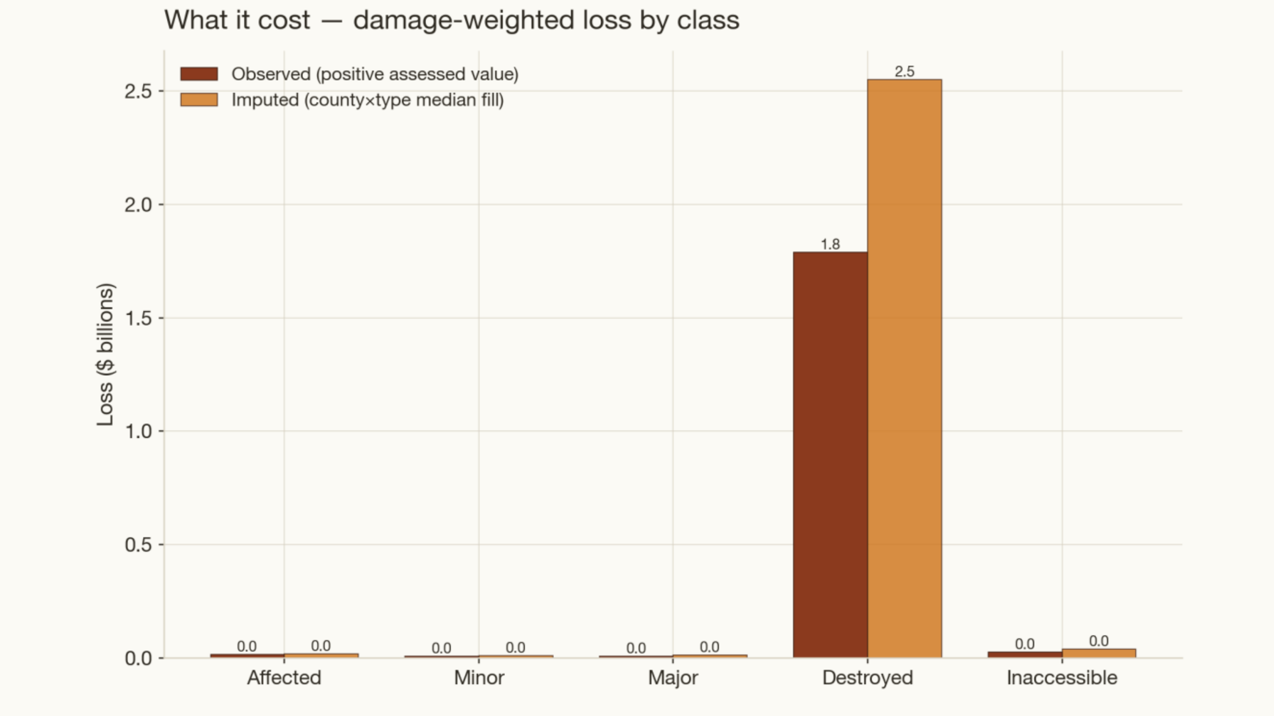
## Calculation

**Base = ASSESSEDIMPROVEDVALUE per structure.**

- damage_fraction weights:
  - Destroyed 1.0
  - Major 0.375
  - Minor 0.175
  - Affected 0.05
  - No Damage 0

## Two reporting approaches

1. **Observed (positive credible values only) = defensible lower bound.**
2. **Imputed (county × structure-type median fill) = coverage-complete estimate.**
   - **Winsorised** *(sensitivity check)* -- cap at 5M$ (see Act 4)




Headline range $1.8B-$2.6B. Assessed improved value understates replacement cost (Prop 13); observed is a defensible lower bound. Sources: CAL FIRE FRAP perimeters & DINS, USFS FVEG/WHR13, USGS MTBS, US Census.

# The season in numbers

**504**

Fires. Dissolved burned footprint 4,180,914 acres.

**89%**

Increase in monthly acreage burned in Aug + Sept

**Top fires by acreage:**

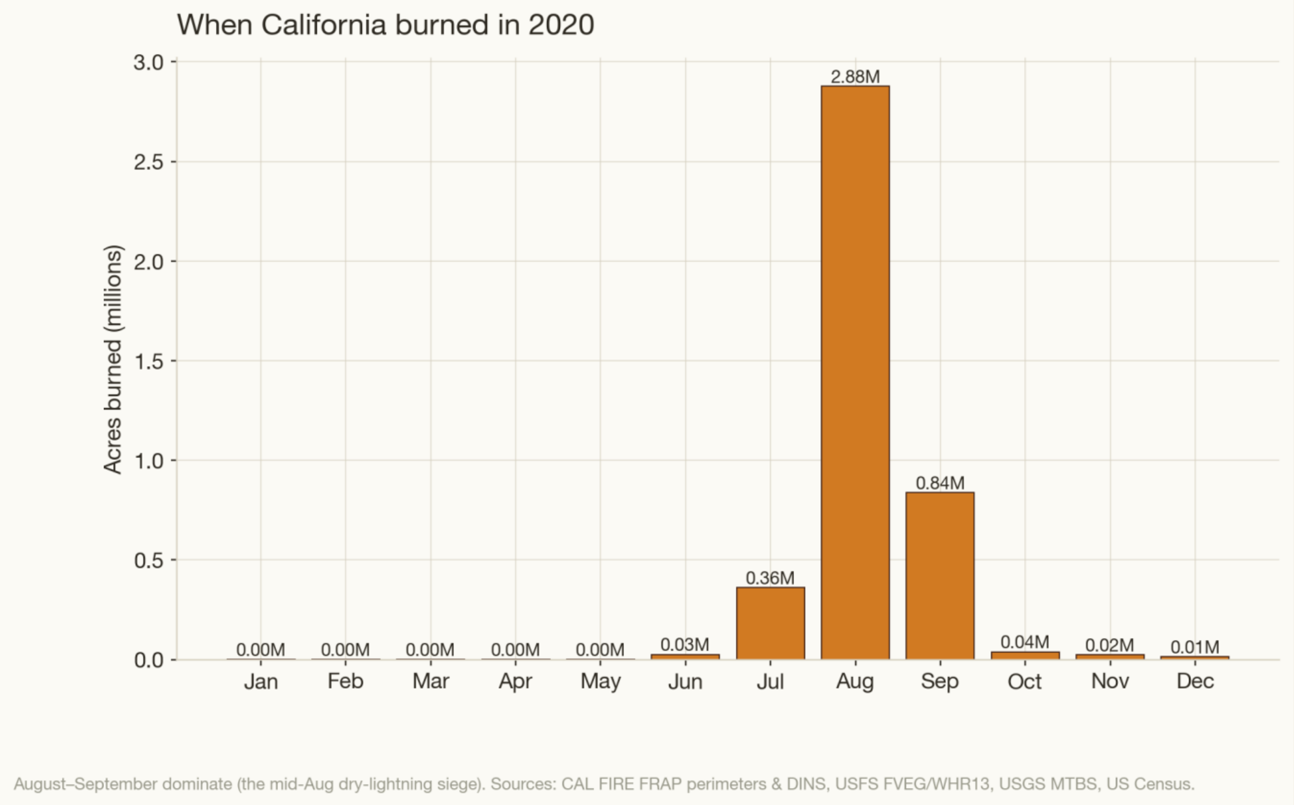
1. **August Complex 1,032,700**
2. **SCU Lightning Complex 396,823**
3. **Creek 379,846**
4. **North Complex 318,799**
5. **Hennessey 305,349**

**Monthly acreage:**

- **Jun 0.026M (107 fires)**
- **Jul 0.36M (116)**
- **Aug 2.88M ac (125)**
- **Sep 0.84M (45)**

Top counties by burned acres:

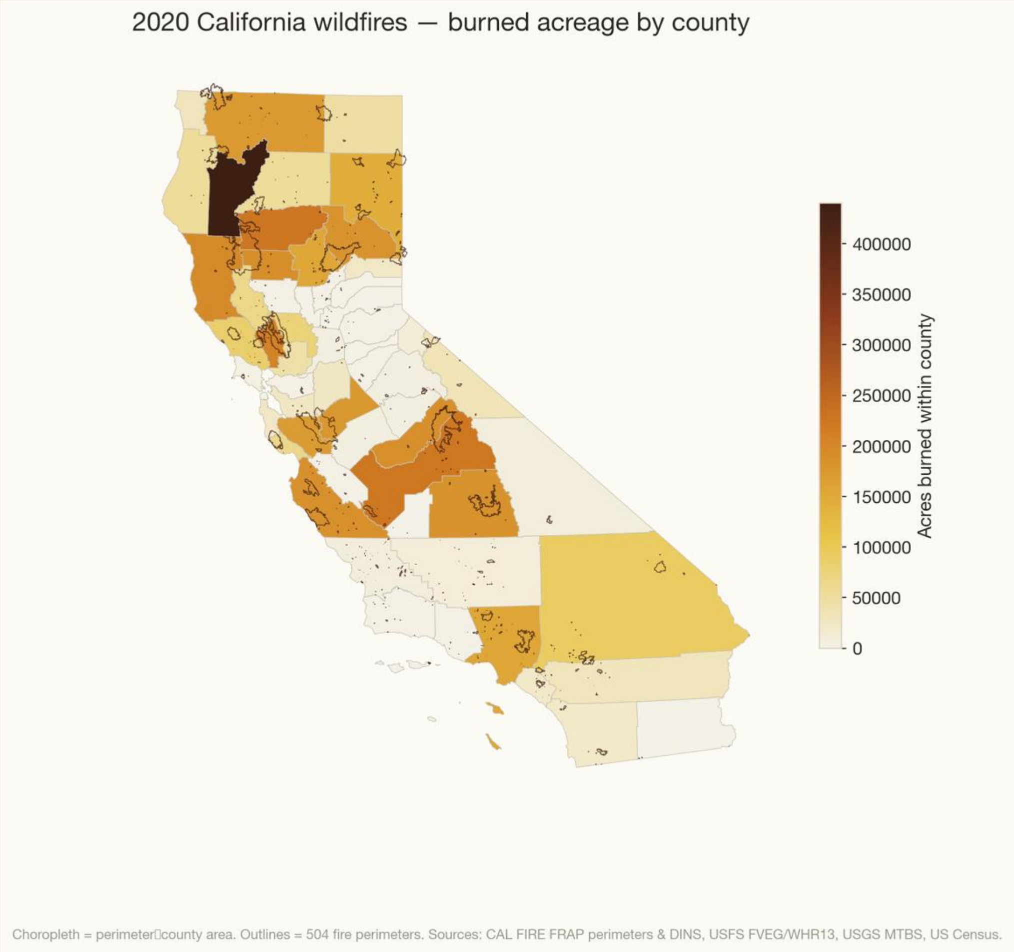
1. Trinity 440k
2. Tehama 224k
3. Fresno 223k
4. Napa 205k
5. Mendocino 195k



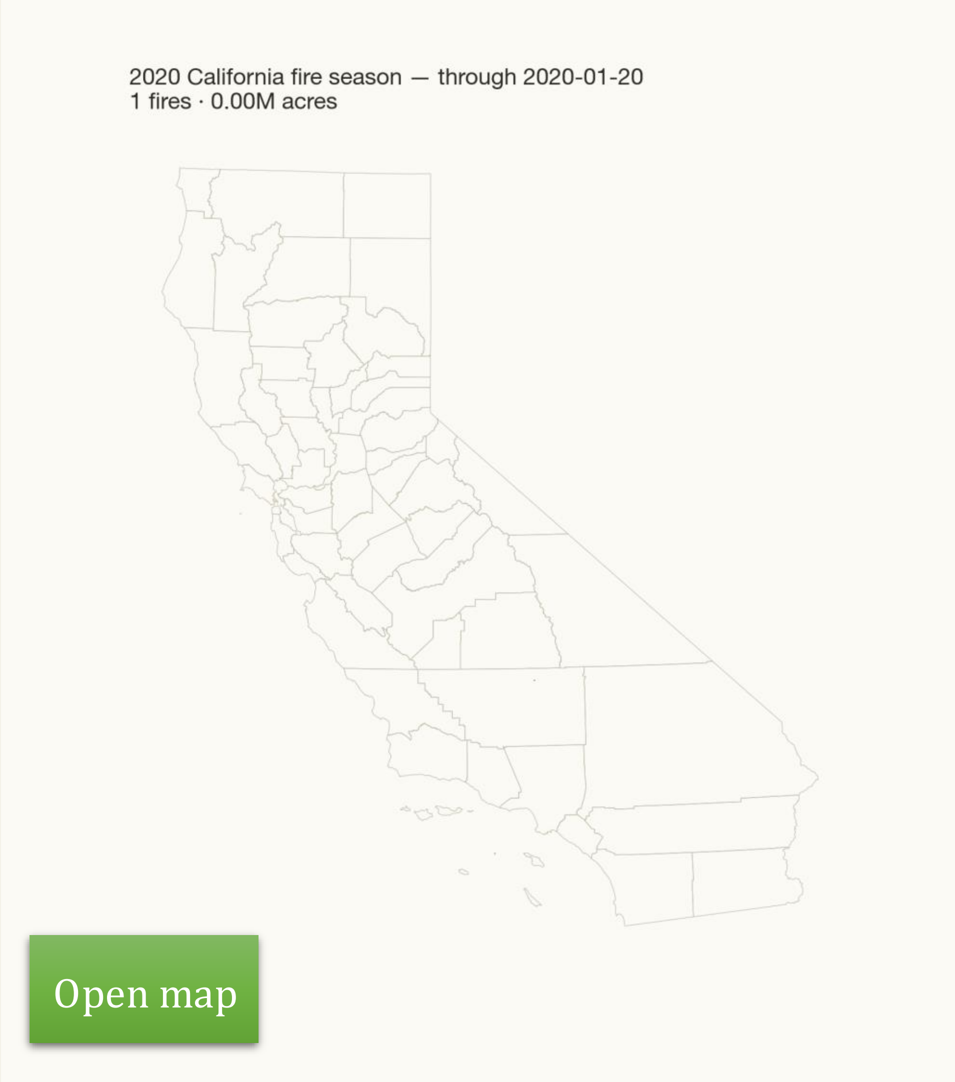

When California burned in 2020 — Aug–Sep ≈ 89% of the season's acreage (the mid-August dry-lightning siege).

ACT 1 · WHERE IT BURNED

# The statewide picture

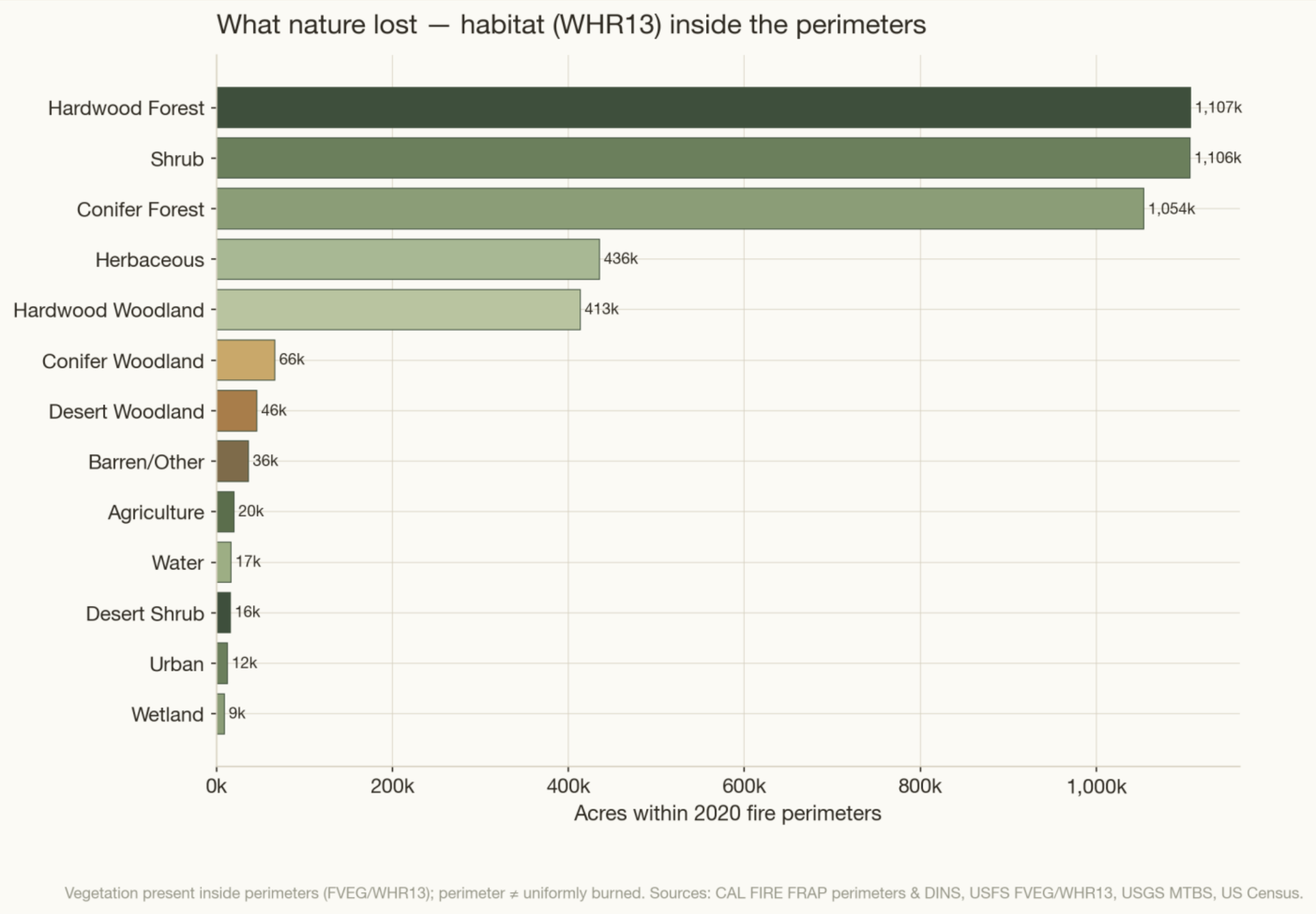



Burned acreage by county with all 504 perimeter outlines. Trinity 440k · Tehama 224k · Fresno 223k · Napa 205k · Mendocino 195k.




The season filling in by alarm week — the August siege. (act1_perimeters_interactive.html)

# Habitat inside the perimeters




Vegetation present within the fire footprint (FVEG, clipped to perimeters then tabulated). A perimeter is NOT uniformly burned.

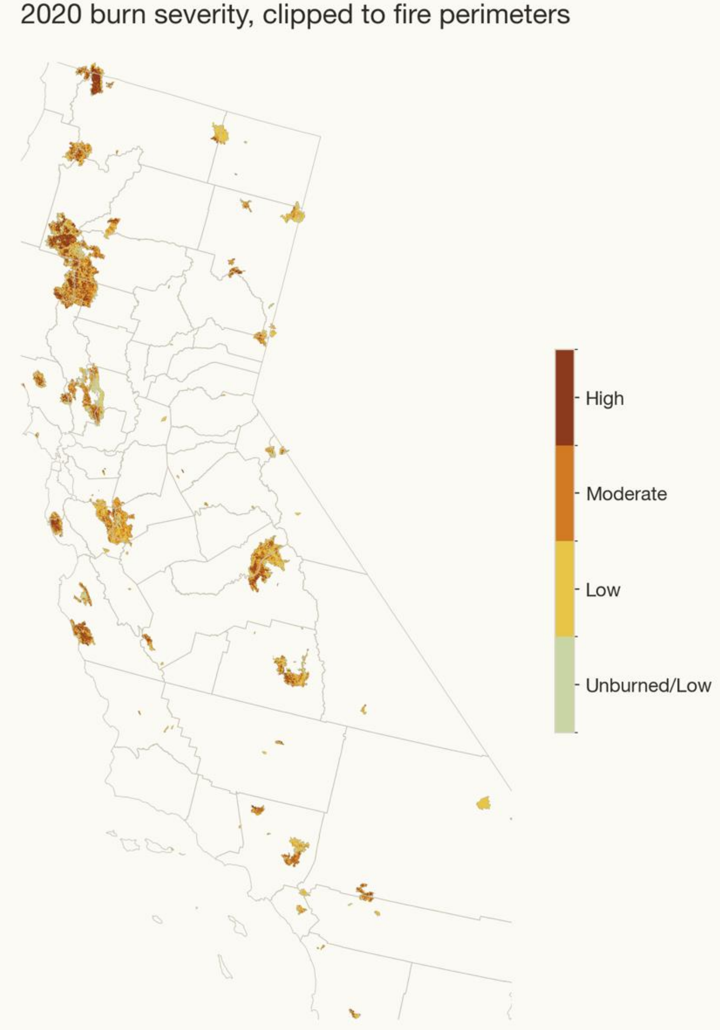
**1,107,352**
Hardwood forest

**1,106,096**
Shrub

**1,053,991**
Conifer forest

ACT 2 · WHAT NATURE LOST

# Burn severity & the coverage caveat



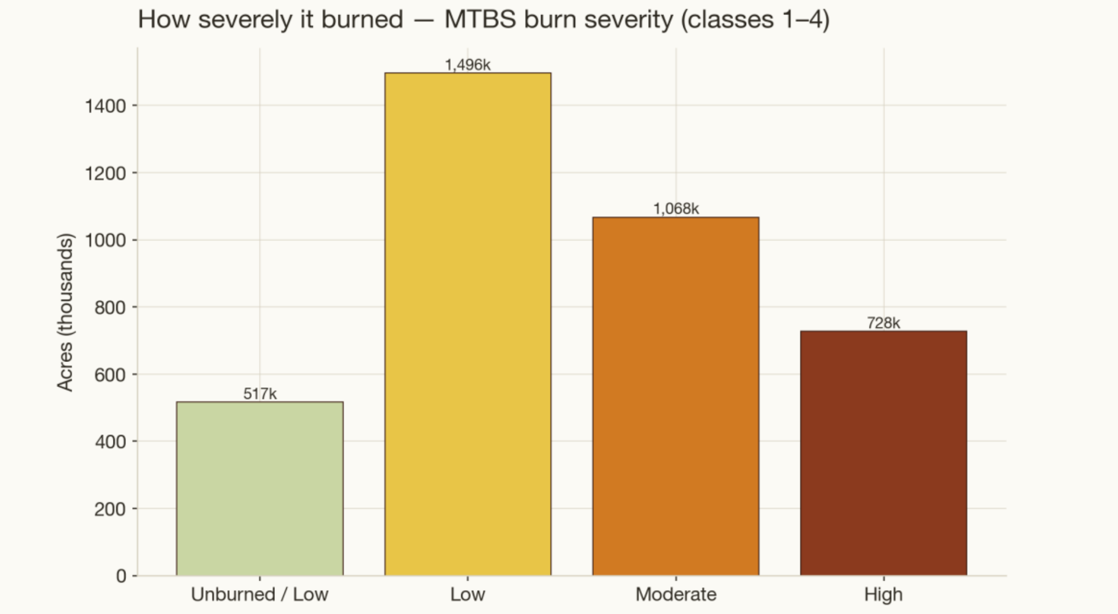

Open map

MTBS thematic severity (classes 1–4). Sources: CAL FIRE FRAP perimeters & DINS, USFS FVEG/WHR13, USGS MTBS, US Census.

**3,808,906**

Total acres burned

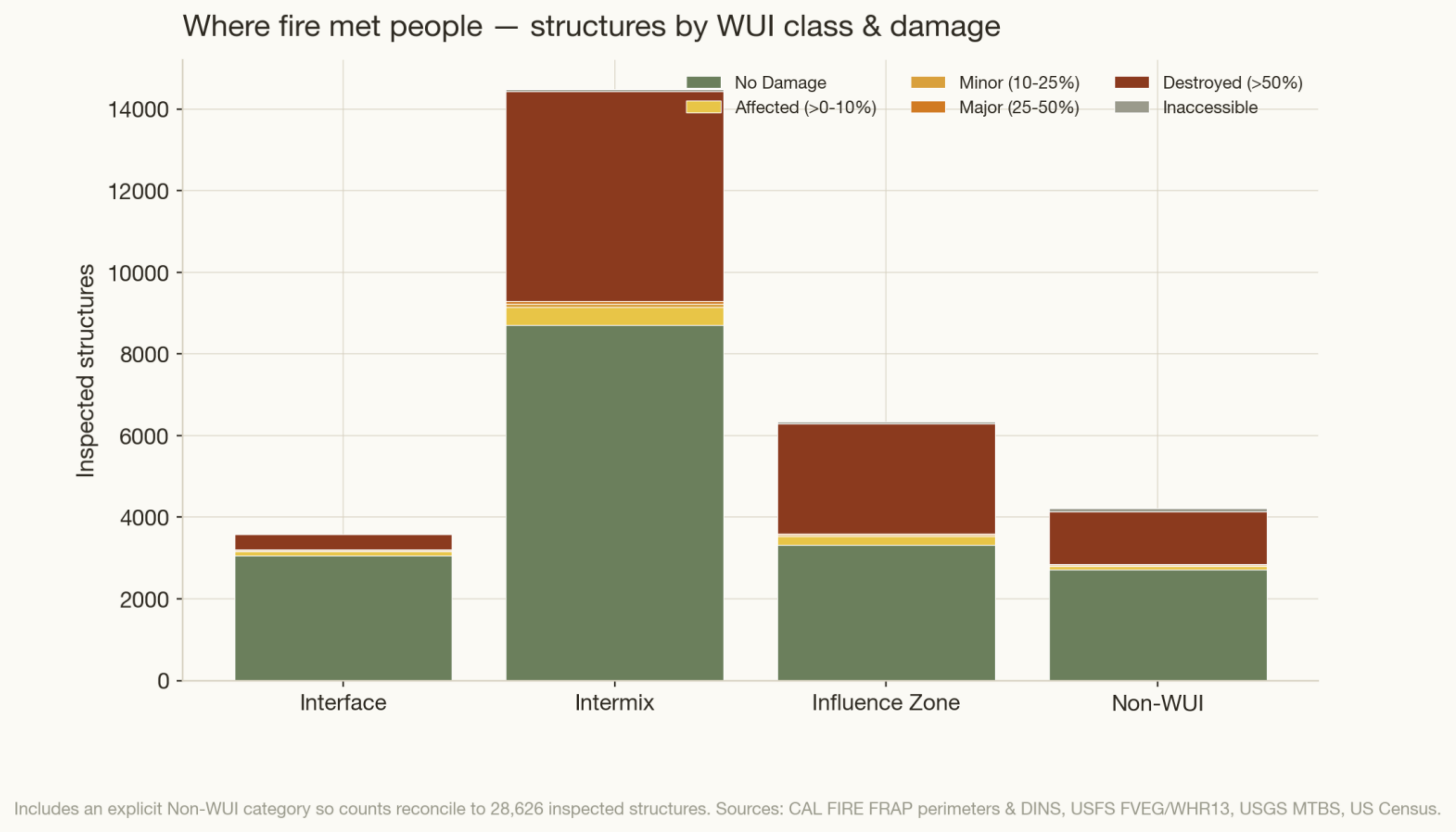
Low 1,496,471 · Moderate 1,067,598 · High 728,187 · Unburned/Low 516,650




MTBS maps fires above ~1,000 acres; this severity analysis covers ~91% of burned acreage (the largest fires), not the long tail of small fires. Increased-greenness (class 5: 8,781 ac) excluded from the burn stack. Sources: CAL FIRE FRAP perimeters & DINS, USFS FVEG/WHR13, USGS MTBS, US Census.

Coverage caveat: MTBS maps fires > ~1,000 acres → covers ~91% of burned acreage, not the small-fire tail (reconciles 3.81M MTBS vs ~4.2M perimeter acres).

# Structures by WUI class & damage




Includes an explicit Non-WUI category so counts reconcile to all 28,626 inspected structures. WUI is a county-level pattern dataset (recent housing vintage), not 2020-exact.

**5,140 / 14,470** ***(36%)***
Intermix structures destroyed

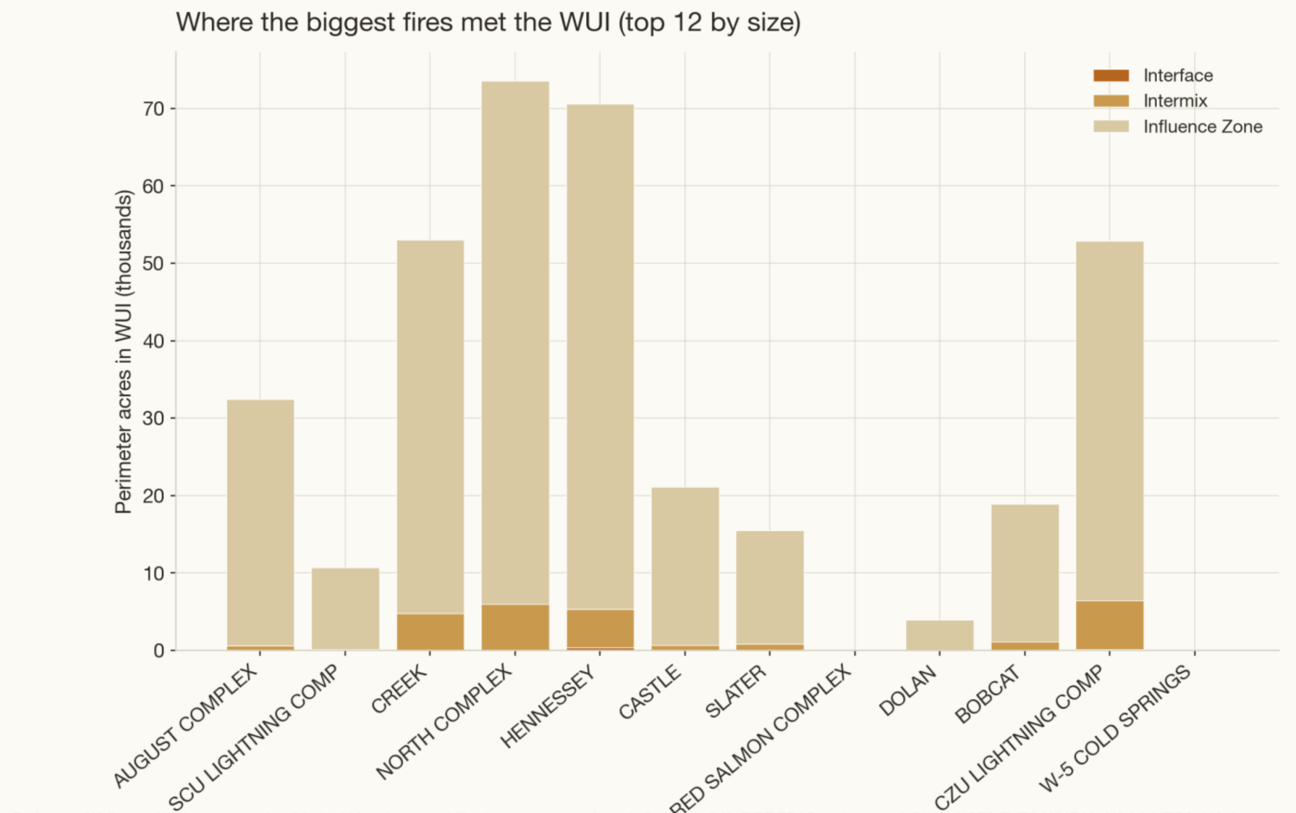
**2,699 / 6,343** ***(43%)***
Influence zone destroyed

**376 / 3,581** ***(10%)***
Interface structures destroyed

- Non-WUI 4,223 — destroyed 1,279 (explicit category, not missing data)

**Take-away: the Intermix (scattered houses within wildland) carries the most inspected structures and the most destruction — the mitigation priority.**

# Where the biggest fires met the WUI




Perimeter∩WUI area; WUI geometry simplified ~100 m for tractability (acreage approximate). Sources: CAL FIRE FRAP perimeters & DINS, USFS FVEG/WHR13, USGS MTBS, US Census.

**14%**

Total burned footprint


- **Statewide perimeter ∩ WUI ≈ 587,435 ac (14.0% of footprint):**
  - Influence Zone 547,493 · Intermix 37,935 · Interface 2,007
- **Per-fire WUI overlap (top 12) :**
  - North Complex 67.6k Influence + 6.0k Intermix
  - Hennessey 65.3k + 5.0k
  - CZU 46.5k + 6.3k
  - Creek 48.3k + 4.8k

# A data defect we had to fix first

## The error

**Raw ASSESSEDIMPROVEDVALUE summed to $30.0B of positive value — implausible.**

- Cause: parcel-aggregate / sentinel artifacts — improbable values repeated across many structures:
  - $54,139,901 on 258
  - $48,703,108 on 45
  - $76,315,906 on 22
  - round $5,000,000 on 109

## The solution

- **Filter: flag a positive value as suspect if > $5M ceiling OR an improbable constant repeated ≥10× above $2M. 849 flagged (75% of dollar value removed; credible positive value = $7.53B).**
  - Audit: 77% (652) caught by the repeated-constant rule; of the rest only 9 are singleton uniques; adding every borderline $5–8M flag back moves the floor by < $0.05B → exclusion errs toward understatement (correct for a floor).

# Damage summary & value coverage

| Damage class | Count | Median credible value | % with positive value |
|---|---|---|---|
| **Destroyed (>50%)** | **9,494** | **$138,348** | **78.9%** |
| No Damage | 17,801 | $230,420 | 81.7% |
| Affected (>0–10%) | 826 | $274,732 | 85.8% |
| Minor (10–25%) | 194 | $203,465 | 75.3% |
| Major (25–50%) | 107 | $186,717 | 63.6% |
| Inaccessible | 204 | $109,890 | 57.4% |

*Destroyed is the loss-bearing class; supports the valuation on the next slide. (Table slide — no new figure.)*

# The headline valuation — a range

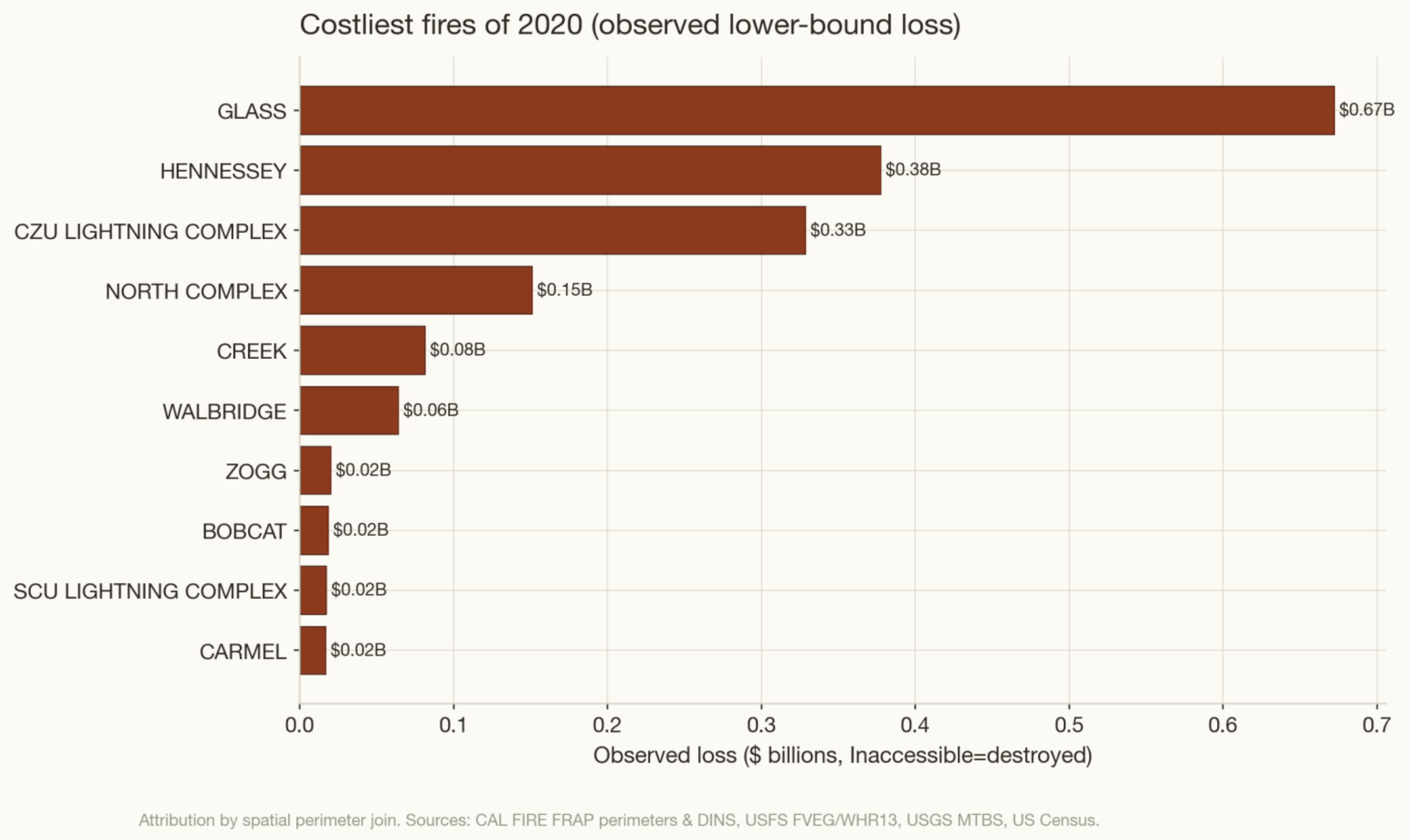
## $1.8 – 2.6 billion

*Assessed-improved-value lower bound for DINS-inspected structures — not total economic loss.*

| $1.82B | $1.85B | $2.63B | $3.62B |
|---|---|---|---|
| Observed lower bound (Inacc. excluded) | Observed (Inacc. = destroyed) | Imputed (county × type median fill) | Winsorised sensitivity (suspects capped $5M) |


Reported as a range with the methodology visible — never a single false-precision number. Assessed values understate replacement cost (Prop 13).

# The costliest fires & the data-selected hotspot : Glass Fire



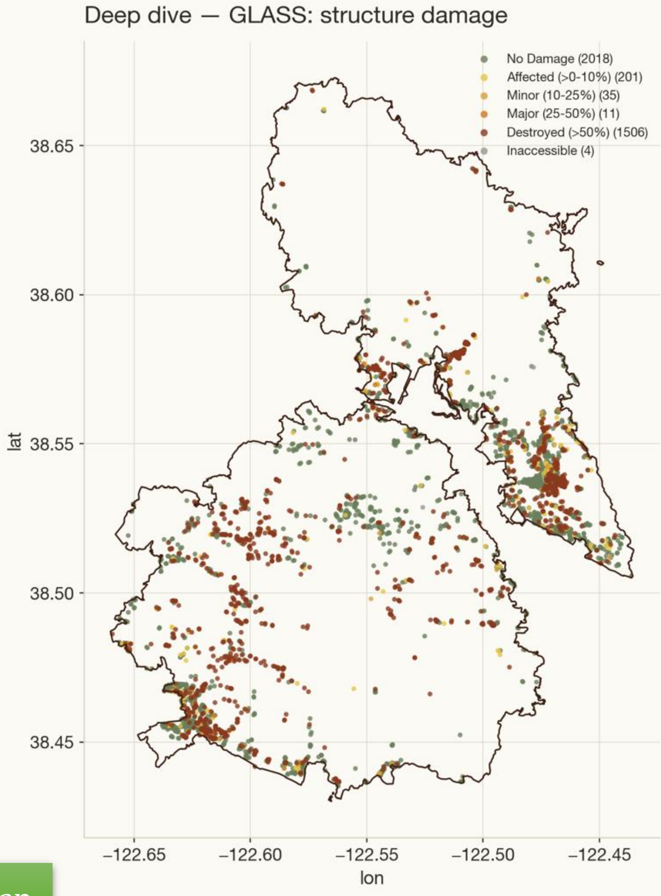

**0.58**
Valuation score

**$0.67B**
Observed loss cost

- **Hotspot score = destroyed count × median credible value:**
  - **Glass 1,506 × $388k = 0.58**
  - Hennessey 1,177 × $315k = 0.37
  - CZU 1,477 × $204k = 0.30
  - North Complex 2,352 × $61k = 0.14
- **High property values + 1,506 destroyed make Glass the data-selected hotspot — exactly the Wine Country cluster anticipated.**

ACT 4 · HOTSPOT DEEP-DIVE

# The Glass fire (Napa / Sonoma)


Open map

DINS inspected structures coloured by damage class. Sources: CAL FIRE FRAP perimeters & DINS, USFS FVEG/WHR13, USGS MTBS, US Census.

**3,775**
inspected

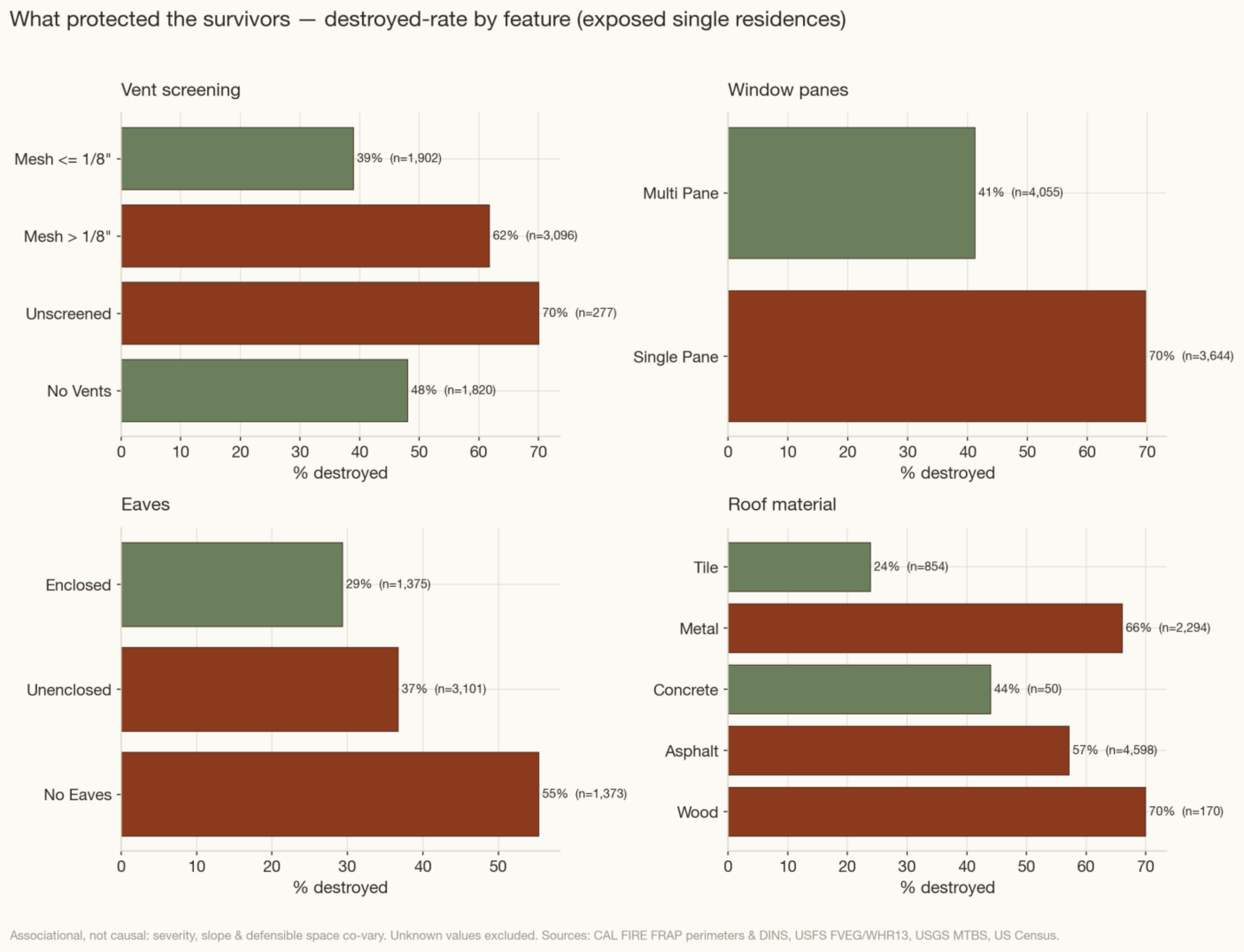
**1,506**
destroyed

**$0.67B**
observed loss

DINS inspected structures within the Glass perimeter, coloured by damage class. 1,506 destroyed, 2,018 no damage; 613 value artifacts excluded.

Also: act4_hotspot_interactive.html — per-structure popups (damage, type, credible value, community).

# What protected the survivors — the empirical hook




**Destroyed-rate among 12,248 exposed single residences, by feature:**

- Vent screen: ≤⅛" mesh 39.0% vs coarse 61.8% vs unscreened 70.0%
- Windows: multi-pane 41.2% vs single-pane 69.8%
- Eaves: enclosed 29.4% vs unenclosed 36.7% vs none 55.4%
- Roof: tile 23.9% vs asphalt 57.1% vs metal 66.1% vs wood 70.0%

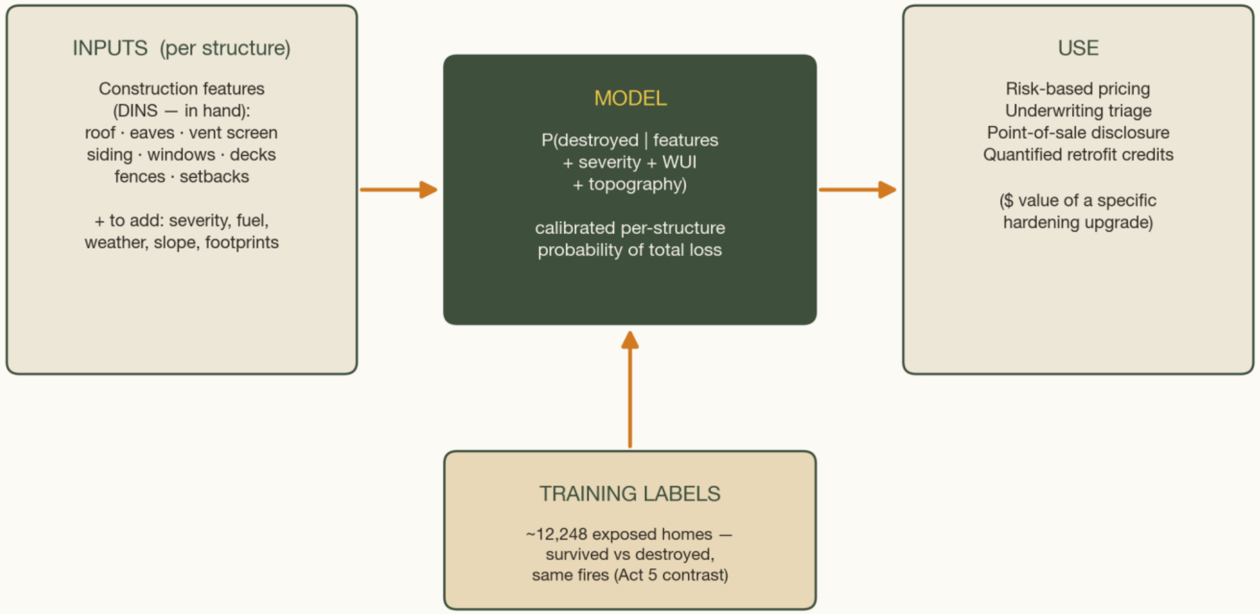
- **Associational, not causal (severity / slope / defensible space co-vary) — but the direction matches CAL FIRE home-hardening guidance.**

# Insurance & a predictive roadmap

**Target** P(destroyed | construction features + severity + WUI class + topography)

1. **Features in hand (DINS): roof, eaves, vent screen, siding, windows, decks, fences, setbacks.**
2. **Data to add: MTBS / FVEG sampled to points; assessor join via APN; replacement-cost models; weather / fuel / topography; building footprints.**
3. **Use: risk-based pricing, underwriting triage, point-of-sale disclosure, quantified retrofit credits.**

From damage record to forward-looking structure-risk model




Roadmap (not a fitted model): DINS construction features + hazard context, trained on the survivor/destroyed contrast, yield a per-structure P(destroyed) for pricing, triage, disclosure and retrofit credits. Sources: CAL FIRE DINS; USGS MTBS; CAL FIRE FVEG.

# Mitigation & the replacement-cost gap

## Mitigation

- Prioritise hardening + defensible space in high-WUI-Intermix × high-severity × high-value zones (Intermix held 5,140 of the destroyed).
- **Lead with the cheap wins: vent mesh, enclosed eaves, multi-pane, class-A roof.**
- Use the survivor map for community outreach.

## The scale-up path

- **Our $1.8–2.6B is an assessed-value floor. Three gaps lift it to true exposure:**
  - assessed → replacement (Prop 13)
  - inspected → complete
  - structures → total economic loss (contents, infrastructure, suppression, business interruption)
- **The assessor APN join is the next increment.**

ONE HONEST SLIDE

# Limitations & caveats

DINS = inspection set, not a census.

Assessed ≠ replacement ≠ market value → headline is a lower bound.

Assessed-value artifacts detected & neutralised (849; 77% repeated-constant).

Perimeter vs DINS naming differs (44% label mismatch; spatial join used).

MTBS maps fires > ~1,000 ac → 91% of burned acreage covered.

WUI = county-level pattern, recent housing vintage, not 2020-exact.

FVEG = vegetation present; perimeter ≠ uniformly burned.

damage_fraction weights are a transparent convention, not appraisal.

# APPENDIX

*Reproducibility · full tables · the artifact filter · asset index*

# A1. METHODOLOGY DETAILS

# Raster clips & MTBS coverage

## Steps to clip

**1. Dissolved + 100 m-buffered mask = 556 non-overlapping parts, 4,412,395 ac (vs 4,182,280 ac summed over individual perimeters — overlapping-complex area correctly removed).**

556

non-overlapping mask parts

4.41M ac

buffered mask area

## MTBS coverage

**91 of 504 perimeters contain ≥1 burned pixel, representing 91.3% of total burned acres.**

- The 413 uncovered perimeters average ~880 ac each — below the ~1,000-acre MTBS threshold, as expected.

91 / 504

perimeters with MTBS burned pixels = 91.3% of burned acres

## Processed outputs

- cleaned DINS / perimeters / WUI / counties (GeoParquet) + FVEG & MTBS tabulations (CSV) + clipped MTBS mosaic (GeoTIFF).

PREP RESULTS · NOTEBOOK 01

# Cleaning & integrity

- **DINS: all 28,626 points have valid geometry; 0 disagree with LATITUDE/LONGITUDE (>1e-5°) → provided geometry trusted.**
- DAMAGE distribution: No Damage 17,801 · Destroyed 9,494 · Affected 826 · Inaccessible 204 · Minor 194 · Major 107.
- Value coverage: 90.5% non-null, 80.6% positive; median positive value $199,962.
- **Perimeters: geometry-derived acres vs reported GIS_ACRES — median ratio 1.00, correlation > 0.99; dates parsed for the time-lapse.**

**0**

geometry vs lat/lon disagreements

**1.00**

acres ratio (geom ÷ reported)

**90.5%**

assessed value non-null

**$200k**

median positive value

METHODOLOGY

# Data-quality guardrails

### Loud failure

Zero-row joins / empty clips raise immediately; shapes & CRS printed at every step.

### Artifact detection

On assessed value (detailed on slide 19) — the single most consequential cleaning step.

### Reproducible

Fixed seed, persisted intermediates, regeneration scripts.

METHODOLOGY

# CRS discipline & clip-first — why the numbers are trustworthy

- All area / distance / join math in EPSG:3310 (California Albers, equal-area); EPSG:4326 used for web display only. CRS asserted before every spatial op.
- Inputs span 4 different CRSs (3857, 4326, 4269, + raster-native 3310 / 5070) — all explicitly reconciled.
- **Clip-first rule: the statewide FVEG (~1.5-billion-pixel) and MTBS rasters are masked to the dissolved + 100 m-buffered perimeters before any tabulation.**
- Windowed reads on non-overlapping parts — correct totals, no double-counting, memory-safe.

**EPSG:3310**

equal-area math CRS

**4 CRSs**

reconciled across inputs

**Clip-first**

rasters masked to perimeters before tabulation

# A2. RASTER CLASS DETAILS

APPENDIX A2

# Full habitat (WHR13) & severity (MTBS) tables

| WHR13 habitat | Acres |
|---|---|
| Hardwood Forest | 1,107,352 |
| Shrub | 1,106,096 |
| Conifer Forest | 1,053,991 |
| Herbaceous | 435,530 |
| Hardwood Woodland | 413,436 |
| Conifer Woodland | 66,116 |
| Desert Woodland | 46,163 |
| Barren / Other | 36,370 |
| Agriculture | 20,115 |
| Water | 16,742 |
| Desert Shrub | 15,793 |
| Urban | 12,252 |
| Wetland | 8,924 |

| MTBS class | Acres | Role |
|---|---|---|
| 1 Unburned/Low | 516,650 | burned |
| 2 Low | 1,496,471 | burned |
| 3 Moderate | 1,067,598 | burned |
| 4 High | 728,187 | burned |
| **Burned total (1–4)** | **3,808,906** | — |
| 5 Increased Greenness | 8,781 | separate |
| 6 Non-Processing Mask | 3,273 | excluded |